

Fatma Zayed

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LinkedIn | GitHub

PROFILE

AI Data Engineer specializing in Machine Learning, Deep Learning, NLP, and Large Language Models (LLMs). Experienced in designing and developing end-to-end AI solutions, including data preprocessing, feature engineering, model training, and deployment. Built real-world projects such as AI-powered data analysis platforms and intelligent multi-agent systems, with a strong focus on automation and practical impact.

EDUCATION

King Salman International University — Faculty of Computer Science and Engineering (AI Department) Sep 2020 – Jun 2025
Bachelor's Degree in Artificial Intelligence Engineering Grade: Very Good
El-Tur, South Sinai, Egypt

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, SQL, R, JavaScript, HTML/CSS
Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn
Web & App Development: Flask, FastAPI, Streamlit, React, Node.js
AI & Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision, Transformers, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI
LLM & AI Frameworks: Hugging Face Transformers, LangChain, LlamaIndex
Data & Analytics: Exploratory Data Analysis (EDA), Data Visualization, Tableau, Power BI
Tools & Platforms: Git, Docker, Jupyter Notebook, Google Colab, VS Code
Databases & Retrieval: Vector Databases (FAISS, ChromaDB)
Automation & Workflows: AI Automation, Workflow Orchestration (n8n), API Integration

EXPERIENCE

AI Intern Jul 2024 – Sep 2024
Zewail City of Science and Technology *Cairo, Egypt*

- Developed ML models (regression, classification, RL) on real datasets improving prediction accuracy.
- Applied neural networks to enhance performance and efficiency.
- Optimized data preprocessing pipelines and workflows.

AI & Data Trainee Jun 2023 – Aug 2023
AOI Academy *Cairo, Egypt*

- Built and evaluated machine learning models.
- Improved model performance through optimization techniques.

AI Trainee 2024
Information Technology Institute (ITI)

- Trained in AI, data analysis, and Generative AI.
- Developed projects using Python, SQL, and LLMs.

Freelance AI Engineer Jan 2023 – Jun 2024
Remote

- Built AI systems for plant disease detection, sentiment analysis, and fake news classification.
- Developed NLP chatbots using Flask and OpenAI API.
- Integrated CNN and LSTM models into real-time applications.

PROJECTS

Farmer Eye Robotic Car (Graduation Project – A+) Built AI-powered robotic vehicle for crop monitoring and plant disease detection (92% accuracy). Integrated Arduino, Raspberry Pi, IoT sensors, cameras, and Flutter mobile app.	2025
End-to-End Data Pipeline & Dashboard Project Extracted, cleaned, and transformed large datasets using SQL and Python. Built interactive dashboards using Tableau and Power BI.	2025
LLM-Powered Data Analyst Agent Built AI agent converting natural language queries into SQL and visual insights. Integrated LLMs, RAG, and Python data pipelines.	2025
Autonomous AI Company (Multi-Agent System) Developed advanced multi-agent AI system simulating real-world company workflows. Implemented task decomposition, inter-agent communication, and dynamic decision-making. Integrated memory systems, vector databases, and external tools for automation.	2025
AI-Powered Data Exploration Platform Engineered end-to-end AI-driven web platform for dataset upload, preprocessing, and visualization. Built secure authentication and automated data preprocessing pipelines. Integrated AI chatbot for real-time chart explanations and data-driven insights.	2025
Full AI SaaS Platform (AI Automation System) Built full-stack AI SaaS platform with authentication, dashboards, and task management. Designed modular multi-agent architecture for workflow automation and orchestration. Implemented scalable backend APIs, memory systems, and real-time processing pipelines.	2025
AI Chatbot with LangChain & OpenAI API Developed context-aware chatbot using RAG and prompt engineering. Improved response accuracy and performance by 25%.	2024
Fake News Detection Web App Built NLP-based web app using TF-IDF, Logistic Regression, and LSTM. Achieved 95% classification accuracy and deployed with Streamlit.	2024

CERTIFICATIONS

- NVIDIA–ITI Generative AI Engineer Training (35 hrs) — Information Technology Institute (ITI), 2023–2026
- AI for All: From Basics to GenAI Practice — NVIDIA Academy (ITI collaboration), 2025
- Transformers in Computer Vision — Mahara-Tech (ITI), 2023
- Understanding Agentic AI — 2023
- Get Started with AI Agent Development on Azure — Microsoft Learn, 2023
- Artificial Intelligence Machine Learning — Zewail City of Science and Technology, 2024
- Artificial Intelligence — AOI Academy (Huawei Certified), 2023
- Machine Learning — Stanford University (Coursera)
- Deep Learning Specialization — DeepLearning.AI (Coursera)
- Python for Data Science — IBM
- IoT — University of California
- Cloud Computing — Coursera